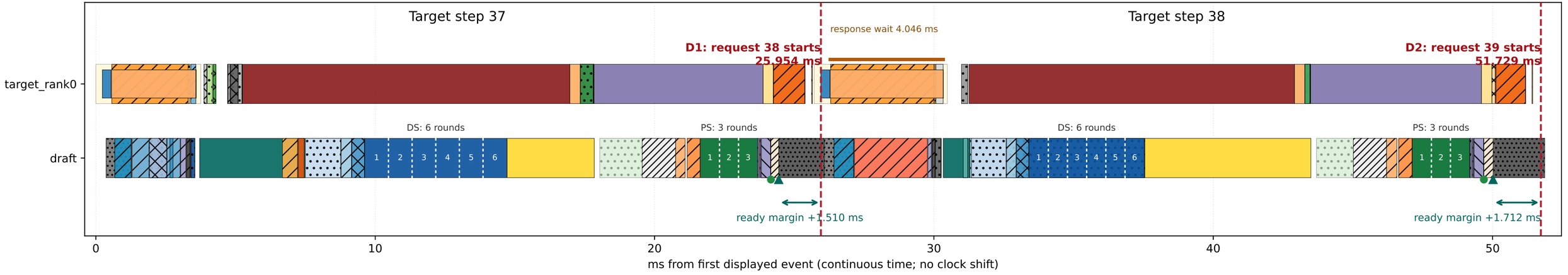
LayerSkip Llama2-7B + llama-68M | Selected setting | P1 hit | UID 288_t0 | steps 37-38

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	115	35.826	2.817	109.397
P2 hit	164	51.090	2.366	93.922
Miss	42	13.084	2.024	79.867
Overall	**321**	**86.916**	**2.483**	**97.707**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	115	25.754	7.104	+3.588	0.830	6.378	+1.638	0.679
P2 hit	164	25.190	6.939	+4.204	0.802	6.428	+1.369	0.681
Miss	42	25.340	6.327	+5.751	0.689	6.211	+1.687	0.680
Overall	**321**	**25.411**	**6.918**	**+4.186**	**0.797**	**6.382**	**+1.507**	**0.680**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

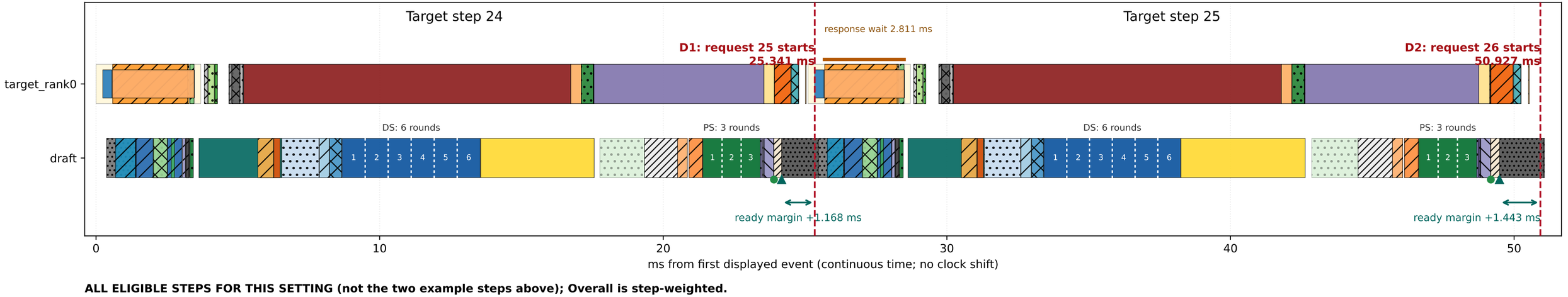
LayerSkip Llama2-7B + llama-68M | Selected setting | P2 hit | UID 244_t0 | steps 24-25

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

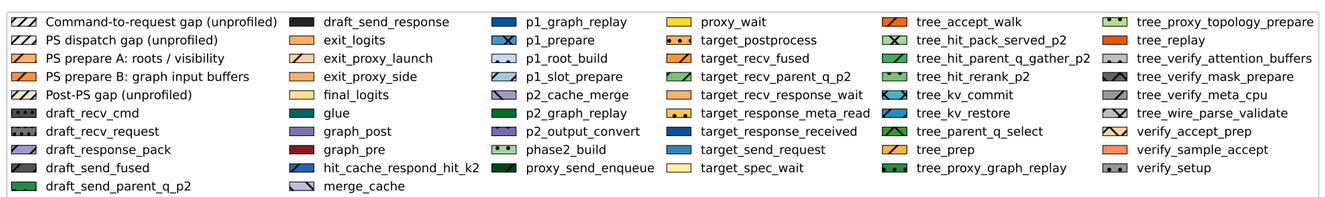
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	115	35.826	2.817	109.397
P2 hit	164	51.090	2.366	93.922
Miss	42	13.084	2.024	79.867
Overall	**321**	**86.916**	**2.483**	**97.707**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
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Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

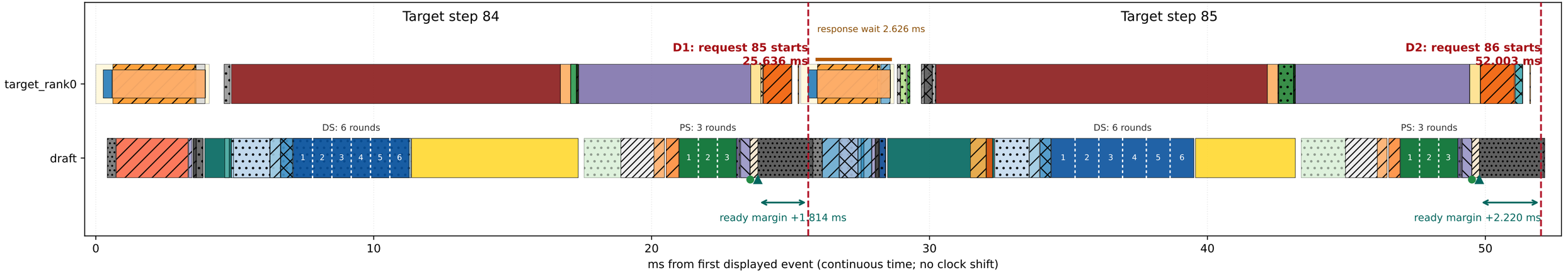
LayerSkip Llama2-7B + llama-68M | Selected setting | Miss | UID 288_t0 | steps 84-85

Historical v1.0 82d964e0 | exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

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Overall	**321**	**25.411**	**6.918**	**+4.186**	**0.797**	**6.382**	**+1.507**	**0.680**

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Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
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chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

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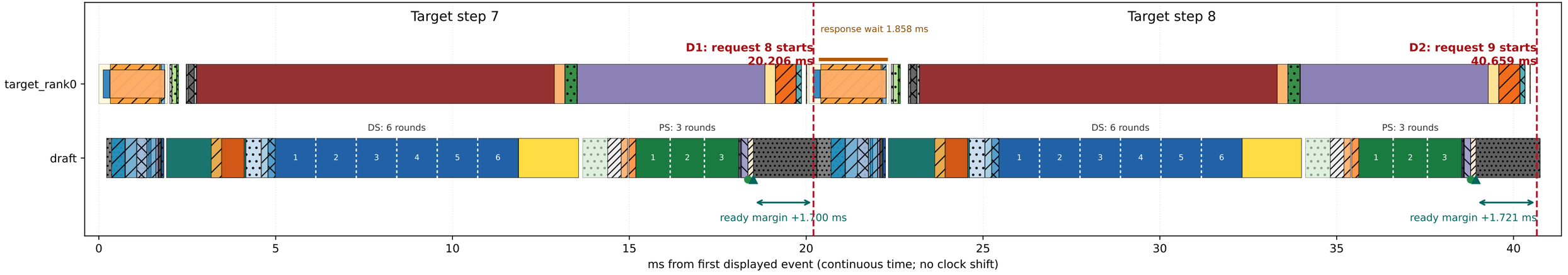
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P1 hit | UID 514_t0 | steps 7-8

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

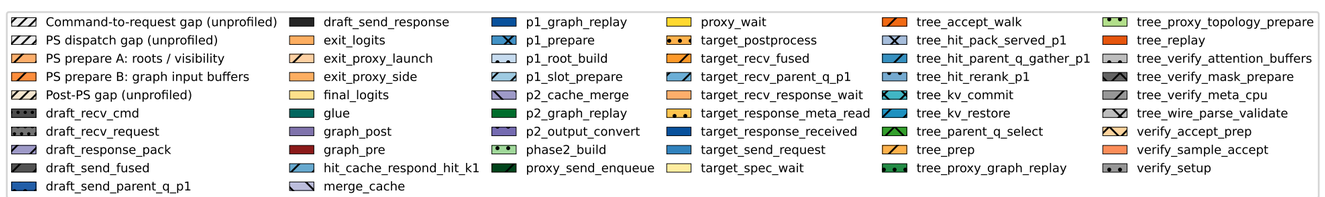
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	813	34.003	3.399	170.993
P2 hit	1245	52.070	2.010	102.241
Miss	333	13.927	1.937	92.621
Overall	**2391**	**86.073**	**2.472**	**124.175**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

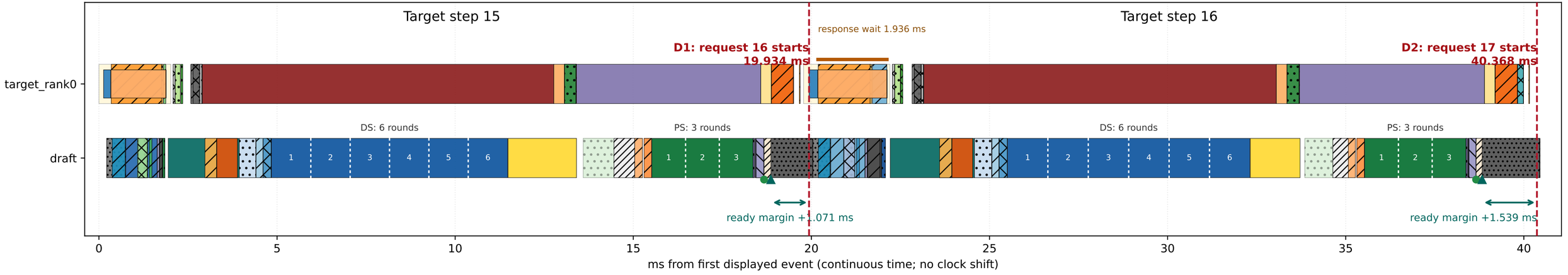
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | P2 hit | UID 397_t0 | steps 15-16

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

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Overall	**2391**	**86.073**	**2.472**	**124.175**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

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Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

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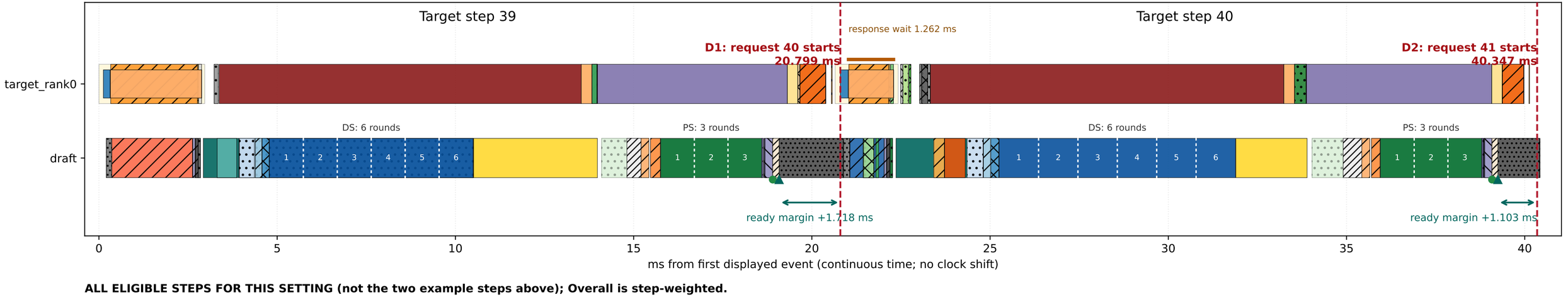
LayerSkip Llama2-7B + AMD-Llama-135M | Adopted PS optimization | Miss | UID 234_t0 | steps 39-40

Runtime matches ver1.2 9390e21 | RTX4090 | B1 BF16 T0.7 | exit20 K1/K2 6/3 M1/M2 12/8 C3 | frontier

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

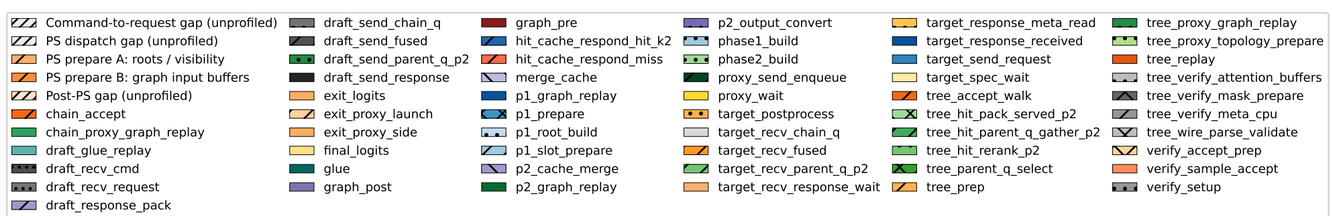
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	813	34.003	3.399	170.993
P2 hit	1245	52.070	2.010	102.241
Miss	333	13.927	1.937	92.621
Overall	**2391**	**86.073**	**2.472**	**124.175**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	813	19.875	7.715	+1.529	1.137	4.914	+1.482	0.953
P2 hit	1245	19.656	7.567	+1.925	1.109	4.943	+1.391	0.953
Miss	333	20.913	6.604	+3.478	0.952	4.905	+1.638	0.947
Overall	**2391**	**19.905**	**7.483**	**+2.007**	**1.097**	**4.928**	**+1.456**	**0.952**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

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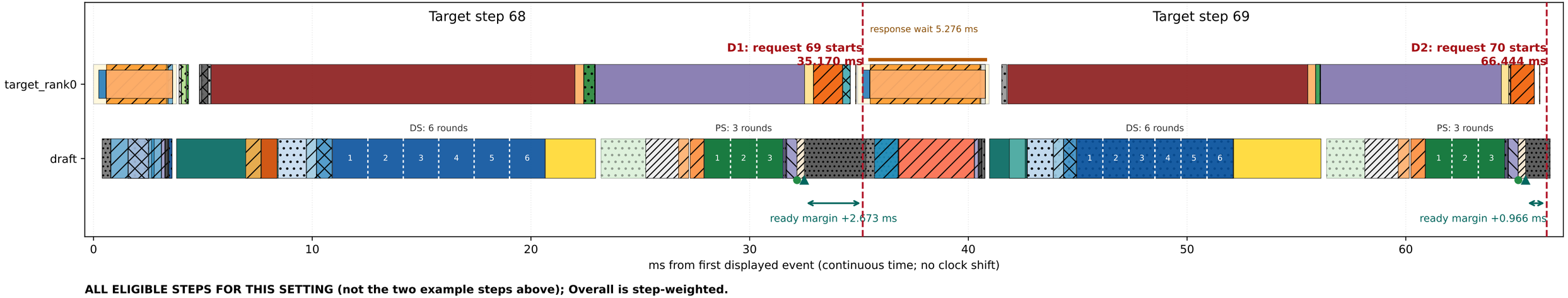
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P1 hit | UID 84_t0 | steps 68-69

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	139	31.306	2.914	81.005
P2 hit	225	50.676	1.871	56.080
Miss	80	18.018	1.675	52.433
Overall	**444**	**81.982**	**2.162**	**63.735**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	139	35.969	12.058	+2.389	1.584	9.352	+1.839	1.208
P2 hit	225	33.365	11.974	+1.955	1.593	9.390	+1.083	1.208
Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

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draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

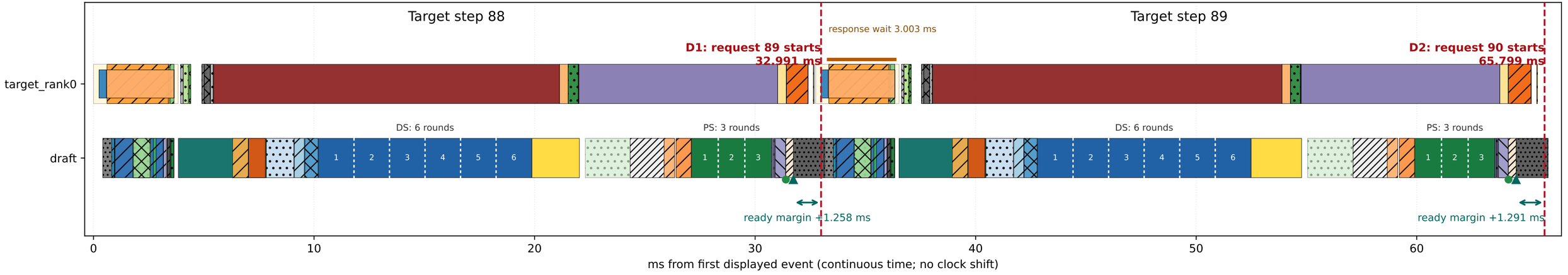
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | P2 hit | UID 244_t0 | steps 88-89

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	139	31.306	2.914	81.005
P2 hit	225	50.676	1.871	56.080
Miss	80	18.018	1.675	52.433
Overall	**444**	**81.982**	**2.162**	**63.735**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	139	35.969	12.058	+2.389	1.584	9.352	+1.839	1.208
P2 hit	225	33.365	11.974	+1.955	1.593	9.390	+1.083	1.208
Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

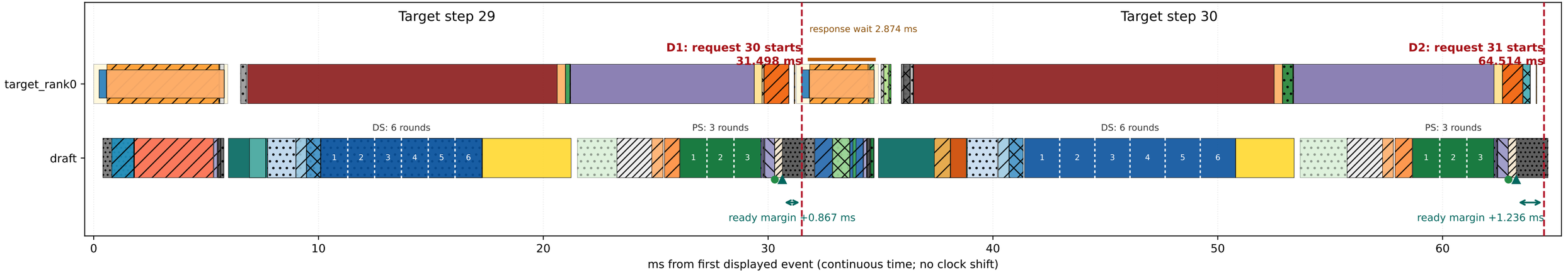
LayerSkip Llama2-13B + AMD-Llama-135M | Selected setting | Miss | UID 128_t0 | steps 29-30

Historical v1.0 59f46511 | exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	139	31.306	2.914	81.005
P2 hit	225	50.676	1.871	56.080
Miss	80	18.018	1.675	52.433
Overall	**444**	**81.982**	**2.162**	**63.735**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
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Miss	80	31.946	9.634	+4.198	1.211	9.095	+0.991	1.209
Overall	**444**	**33.924**	**11.579**	**+2.495**	**1.521**	**9.325**	**+1.303**	**1.208**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

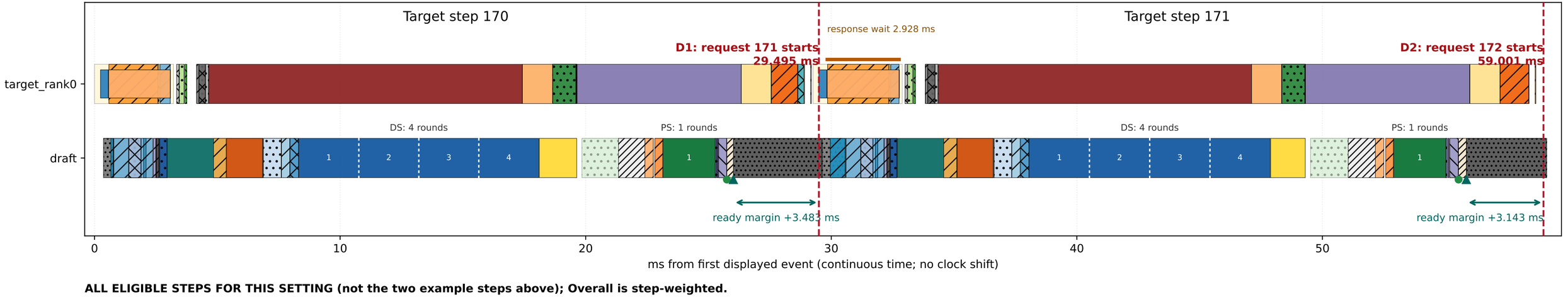
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | P1 hit | UID 84_t0 | steps 170-171

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

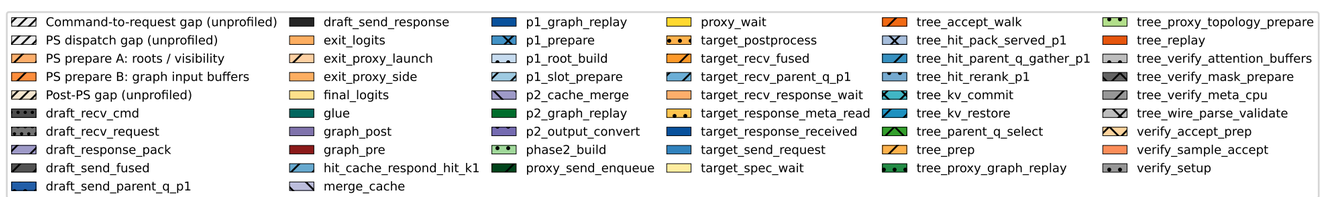
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	65	16.796	1.277	43.679
P2 hit	295	76.227	1.227	43.525
Miss	27	6.977	1.111	39.574
Overall	**387**	**93.023**	**1.227**	**43.279**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	65	29.234	11.203	+1.503	2.419	6.237	+3.108	2.121
P2 hit	295	28.194	10.179	+2.677	2.168	6.257	+2.942	2.122
Miss	27	28.077	9.295	+3.628	1.954	6.069	+3.197	2.126
Overall	**387**	**28.360**	**10.289**	**+2.546**	**2.195**	**6.240**	**+2.987**	**2.122**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

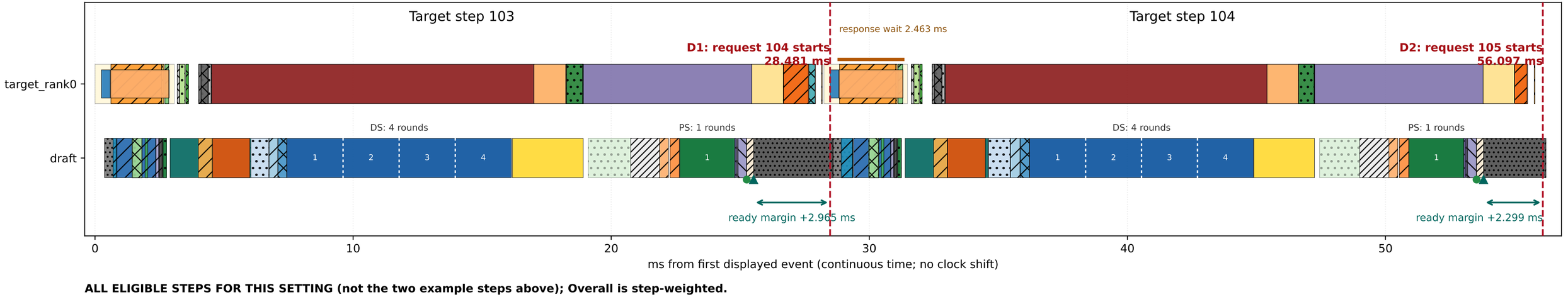
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | P2 hit | UID 84_t0 | steps 103-104

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

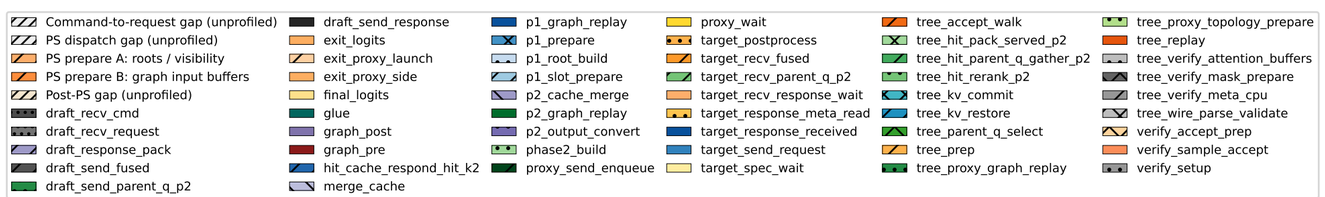
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	65	16.796	1.277	43.679
P2 hit	295	76.227	1.227	43.525
Miss	27	6.977	1.111	39.574
Overall	**387**	**93.023**	**1.227**	**43.279**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

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P1 hit	65	29.234	11.203	+1.503	2.419	6.237	+3.108	2.121
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Miss	27	28.077	9.295	+3.628	1.954	6.069	+3.197	2.126
Overall	**387**	**28.360**	**10.289**	**+2.546**	**2.195**	**6.240**	**+2.987**	**2.122**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.



Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

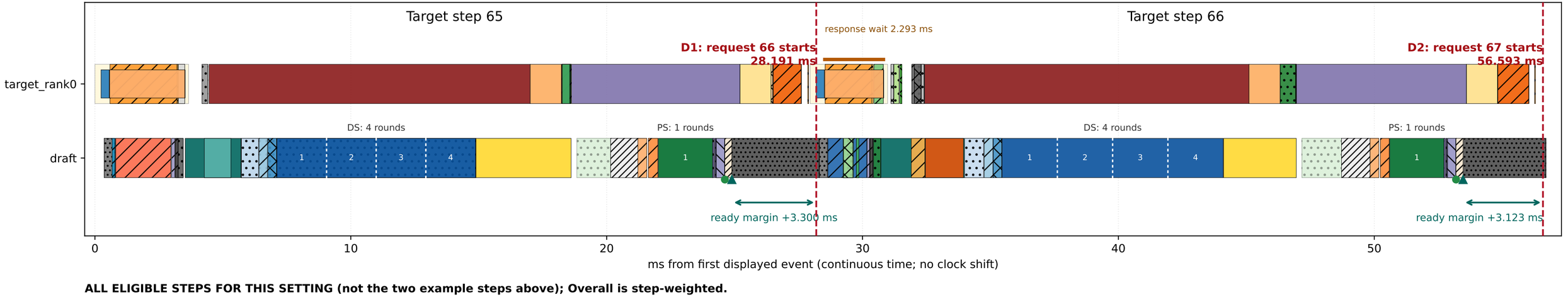
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M | Selected setting | Miss | UID 244_t0 | steps 65-66

Historical v1.0 16dd4198 | exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

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Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
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Miss	27	28.077	9.295	+3.628	1.954	6.069	+3.197	2.126
Overall	**387**	**28.360**	**10.289**	**+2.546**	**2.195**	**6.240**	**+2.987**	**2.122**

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Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k2
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p2
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

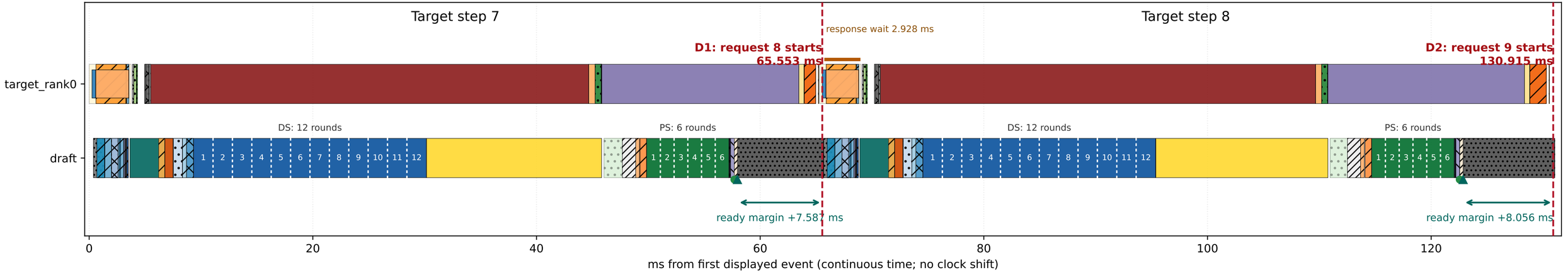
LayerSkip CodeLlama-34B + AMD-Llama-135M-code | Selected setting | P1 hit | UID humaneval_local_033_t0 | steps 7-8

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

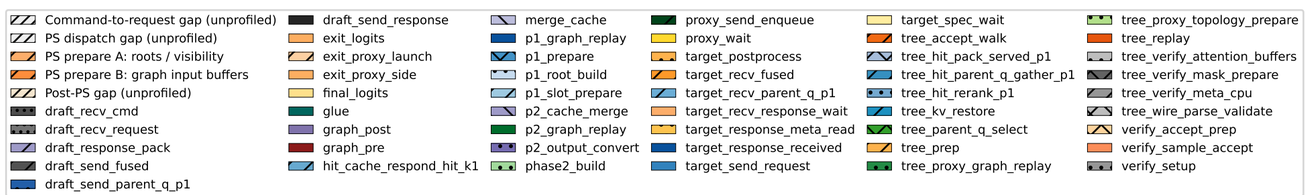
Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	62	84.932	7.290	111.757
P2 hit	11	15.068	5.091	92.013
Miss	0	0.000	N/A	N/A
Overall	**73**	**100.000**	**6.959**	**109.174**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	62	65.234	22.848	+14.976	1.747	11.967	+7.769	1.226
P2 hit	11	55.328	22.743	+8.651	1.743	11.867	+5.450	1.226
Miss	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Overall	**73**	**63.741**	**22.832**	**+14.023**	**1.746**	**11.952**	**+7.420**	**1.226**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.



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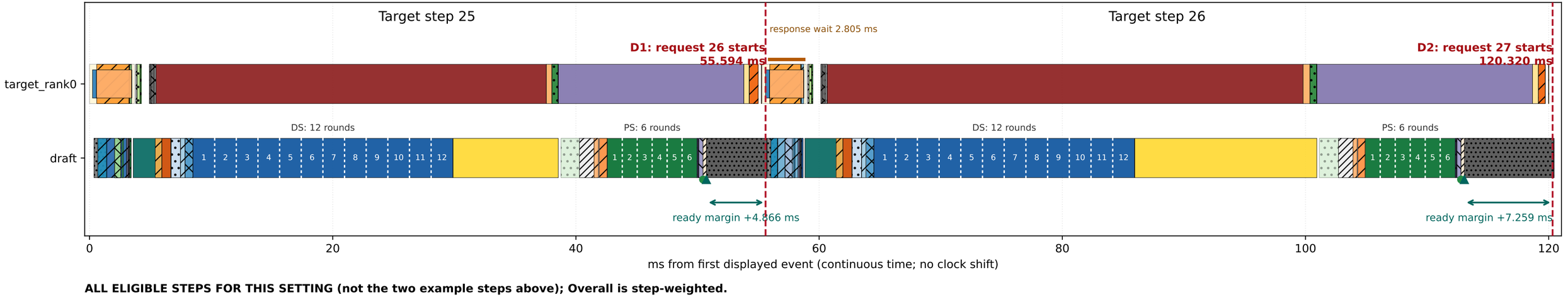
LayerSkip CodeLlama-34B + AMD-Llama-135M-code | Selected setting | P2 hit | UID humaneval_local_023_t0 | steps 25-26

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

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Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
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Miss	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Overall	**73**	**63.741**	**22.832**	**+14.023**	**1.746**	**11.952**	**+7.420**	**1.226**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

Raw intervals retained, no 0.02 ms visibility floor. Cross-device margins inherit CUDA-to-host clock alignment uncertainty; exact raw events and source hashes accompany this figure.

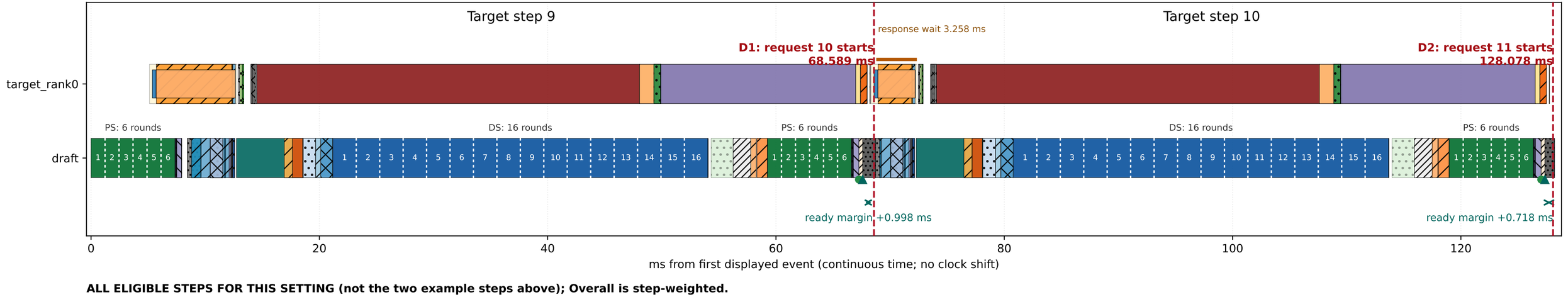
LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | P1 hit | UID humaneval_local_023_t0 | steps 9-10

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

Both target steps and their DS/PS work share one time axis. Only the following request boundary is used for the second deadline.

In this measurement: no observed proxy computation / target graph_post overlap. Send enqueue does not measure transfer completion.



ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	63	75.000	6.508	100.263
P2 hit	19	22.619	4.526	75.642
Miss	2	2.381	1.500	22.657
Overall	**84**	**97.619**	**5.940**	**93.122**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	63	64.908	36.258	-5.479	2.059	14.165	-0.365	1.236
P2 hit	19	59.838	35.969	-6.511	2.058	14.526	-3.691	1.234
Miss	2	66.205	35.801	-8.284	1.867	13.198	-5.588	1.232
Overall	**84**	**63.792**	**36.182**	**-5.779**	**2.054**	**14.224**	**-1.241**	**1.235**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

Hatched gaps = unprofiled elapsed intervals, NOT proven GPU idle. Nested spans are not additive. White graph dividers show counts, NOT per-forward measurements.

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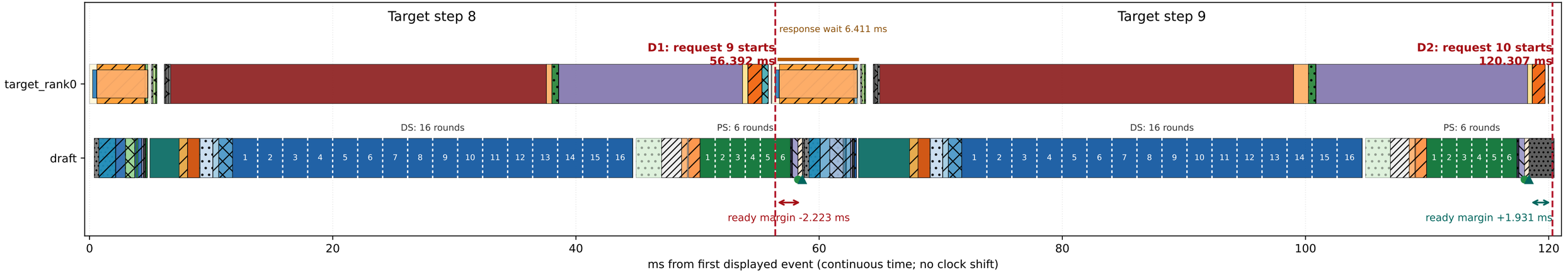
LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | P2 hit | UID humaneval_local_003_t0 | steps 8-9

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Red dashed line = next target request starts (observed deadline reference). Green circle = PS cache ready; triangle = draft ready to receive.

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ALL ELIGIBLE STEPS FOR THIS SETTING (not the two example steps above); Overall is step-weighted.

Current outcome	Steps	Rate (%)	AL (tokens/step)	Profile TPS (tokens/s)
P1 hit	63	75.000	6.508	100.263
P2 hit	19	22.619	4.526	75.642
Miss	2	2.381	1.500	22.657
Overall	**84**	**97.619**	**5.940**	**93.122**

Rate = source / all steps; Overall rate = total hit (P1 + P2) / all. AL includes recovery/bonus. Profile TPS = emitted tokens / summed step seconds.

Current outcome	Steps	Step (ms)	DS total (ms)	DS spare* (ms)	DS graph/K1 (ms)	PS total (ms)	PS spare (ms)	PS graph/K2 (ms)
P1 hit	63	64.908	36.258	-5.479	2.059	14.165	-0.365	1.236
P2 hit	19	59.838	35.969	-6.511	2.058	14.526	-3.691	1.234
Miss	2	66.205	35.801	-8.284	1.867	13.198	-5.588	1.232
Overall	**84**	**63.792**	**36.182**	**-5.779**	**2.054**	**14.224**	**-1.241**	**1.235**

Step = current target request start -> next request start. *DS spare = proxy send-ready - DS end; PS spare = next request start - draft receive-ready.

Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_fused
draft_send_parent_q_p1
draft_send_parent_q_p2
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_hit_k2
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_fused
target_recv_parent_q_p1
target_recv_parent_q_p2
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_pack_served_p2
tree_hit_parent_q_gather_p1
tree_hit_parent_q_gather_p2
tree_hit_rerank_p1
tree_hit_rerank_p2
tree_kv_commit
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
verify_setup

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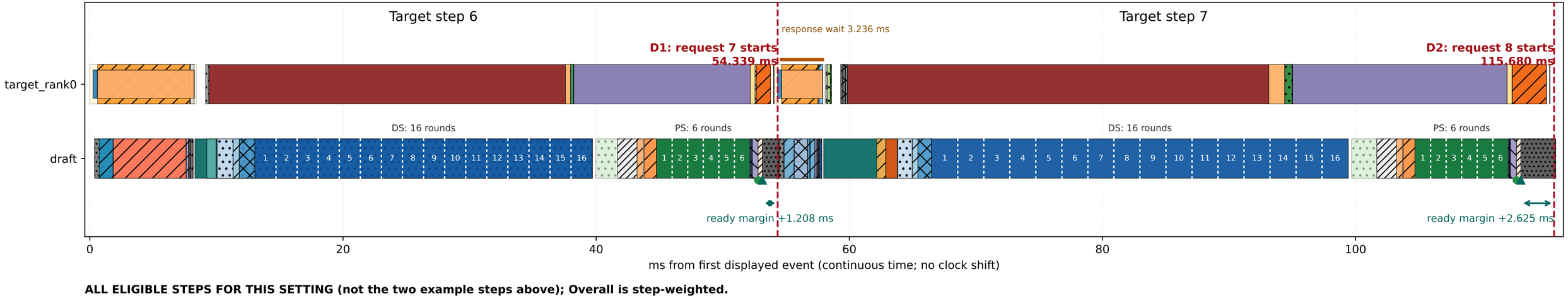
LayerSkip CodeLlama-34B + AMD-Llama-135M-code | K1/K2 16/6 (throughput alternative) | Miss | UID humaneval_local_023_t0 | steps 6-7

Historical v1.0 59f46511 | exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

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Positive spare = early; negative = late. Graph/K includes sampling/selection, not isolated transformer latency. These are profiled-subset TPS, not full evaluation TPS.

Command-to-request gap (unprofiled)
PS dispatch gap (unprofiled)
PS prepare A: roots / visibility
PS prepare B: graph input buffers
Post-PS gap (unprofiled)
chain_accept
chain_proxy_graph_replay
draft_glue_replay
draft_recv_cmd
draft_recv_request
draft_response_pack
draft_send_chain_q
draft_send_fused
draft_send_parent_q_p1
draft_send_response
exit_logits
exit_proxy_launch
exit_proxy_side
final_logits
glue
graph_post
graph_pre
hit_cache_respond_hit_k1
hit_cache_respond_miss
merge_cache
p1_graph_replay
p1_prepare
p1_root_build
p1_slot_prepare
p2_cache_merge
p2_graph_replay
p2_output_convert
phase1_build
phase2_build
proxy_send_enqueue
proxy_wait
target_postprocess
target_recv_chain_q
target_recv_fused
target_recv_parent_q_p1
target_recv_response_wait
target_response_meta_read
target_response_received
target_send_request
target_spec_wait
tree_accept_walk
tree_hit_pack_served_p1
tree_hit_parent_q_gather_p1
tree_hit_rerank_p1
tree_kv_restore
tree_parent_q_select
tree_prep
tree_proxy_graph_replay
tree_proxy_topology_prepare
tree_replay
tree_verify_attention_buffers
tree_verify_mask_prepare
tree_verify_meta_cpu
tree_wire_parse_validate
verify_accept_prep
verify_sample_accept
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